

Day 2 Lectures

AJ Smit

This is a heading

This is the same comment that I hash out inside the code block: load the tidyverse library

```
# load the tidyverse library  
library(tidyverse)
```

```
mean(1, 2, 3)
```

```
[1] 1
```

Numeric data

```
# Create a vector of the number of pets owned by each household  
num_rats <- c(0, 1, 2, 2, 3, 1, 4, 0, 2, 1, 2, 2, 0, 3, 2, 1, 1, 4, 2, 0)  
num_rats
```

```
[1] 0 1 2 2 3 1 4 0 2 1 2 2 0 3 2 1 1 4 2 0
```

```
class(num_rats)
```

```
[1] "numeric"
```

```
num_rats_int <- as.integer(num_rats) # coercion  
num_rats_int
```

```
[1] 0 1 2 2 3 1 4 0 2 1 2 2 0 3 2 1 1 4 2 0
```

```
class(num_rats_int)
```

```
[1] "integer"
```

```
pies <- pi * seq(1:5)
pies
```

```
[1] 3.141593 6.283185 9.424778 12.566371 15.707963
```

```
class(pies)
```

```
[1] "numeric"
```

```
as.integer(pies)
```

```
[1] 3 6 9 12 15
```

```
round(pies, 2)
```

```
[1] 3.14 6.28 9.42 12.57 15.71
```

The `as.integer()` function does not care about rounding, for example:

```
as.integer(round(pies, 2))
```

```
[1] 3 6 9 12 15
```

```
set.seed(13)
# Generate a vector of 50 normally distributed temperature values
temp_data <- round(rnorm(n = 50, mean = 15, sd = 3), 2)
# temp_data <- rnorm(n = 50, mean = 15, sd = 3)
temp_data
```

```
[1] 16.66 14.16 20.33 15.56 18.43 16.25 18.69 15.71 13.90 18.32 11.72 16.39
[13] 10.92 9.43 13.68 14.42 19.19 15.30 14.66 17.11 15.79 20.51 16.07 11.86
[25] 16.86 15.45 10.62 8.92 11.83 12.82 14.98 17.54 13.85 13.42 14.18 13.18
[37] 14.00 14.28 12.41 12.46 15.30 19.77 16.70 19.84 13.59 12.82 11.93 9.19
[49] 15.83 19.23
```

```
length(temp_data)
```

```
[1] 50
```

```
mean(temp_data)
```

```
[1] 14.9212
```

```
sd(temp_data)
```

```
[1] 2.92563
```

```
class(temp_data)
```

```
[1] "numeric"
```

```
names <- c("Fluffy", "Jesse")  
names <- c('Fluffy', 'Jesse')  
class(names)
```

```
[1] "character"
```

Factor data

```
# Generate a vector of vegetation types  
  
vegetation <- sample(c("grassland", "forest", "wetland"), size = 50, replace =  
TRUE)  
  
# View the vegetation data  
vegetation
```

```
[1] "forest"    "wetland"   "grassland" "grassland" "grassland" "wetland"  
[7] "grassland" "wetland"   "grassland" "grassland" "wetland"   "forest"  
[13] "wetland"   "forest"    "forest"    "grassland" "wetland"   "grassland"  
[19] "grassland" "grassland" "forest"    "wetland"   "grassland" "forest"  
[25] "grassland" "forest"    "forest"    "grassland" "forest"    "grassland"  
[31] "grassland" "wetland"   "wetland"   "grassland" "grassland" "forest"  
[37] "wetland"   "grassland" "forest"    "forest"    "grassland" "grassland"  
[43] "forest"    "forest"    "grassland" "forest"    "grassland" "wetland"  
[49] "wetland"   "forest"
```

```
class(vegetation)
```

```
[1] "character"
```

```
veg_fact <- as.factor(vegetation)
class(veg_fact)
```

```
[1] "factor"
```

Ordered factors

```
# Vector of ordinal data representing the successional stage of a forest
coffee <- c("Small", "Tall", "Venti", "Grande", "Ginormous")
class(coffee)
```

```
[1] "character"
```

```
# Convert to ordered factor
coffee <- factor(coffee, ordered = TRUE,
                 levels = c("Small", "Tall", "Venti",
                           "Grande", "Ginormous"))
coffee
```

```
[1] Small    Tall     Venti    Grande   Ginormous
Levels: Small < Tall < Venti < Grande < Ginormous
```

```
class(coffee)
```

```
[1] "ordered" "factor"
```

Logical data

```
# Generate a vector of 1s and 0s to represent the presence
# or absence of a species in different ecological sites
species_presence <- sample(c(0, 1), 10,
                          replace = TRUE)
class(species_presence)
```

```
[1] "numeric"
```

```
species_presence_logi <- as.logical(species_presence)
class(species_presence_logi)
```

```
[1] "logical"
```

Dates

```
my_time <- as.POSIXct("2022-03-10 12:34:56")
class(my_time)
```

```
[1] "POSIXct" "POSIXt"
```

Missing values

```
# Set the length of the sequence
n <- 100

# Generate a sequence of random normal numbers with
# mean 0 and standard deviation 1
data <- rnorm(n, mean = 0, sd = 1)

# Randomly assign 5% of the values as missing
missing_indices <- sample(1:n, size = round(0.05*n))
data[missing_indices] <- NA
length(data)
```

```
[1] 100
```

```
mean(data, na.rm = TRUE)
```

```
[1] -0.1332528
```

```
length(data)
```

```
[1] 100
```

```
data_sans_na <- na.omit(data)
length(data_sans_na)
```

```
[1] 95
```

Matrices (plural) and a matrix (singular)

```
# create a numeric matrix
my_matrix_1 <- matrix(1:6, nrow = 2, ncol = 3,
                     byrow = FALSE)

my_matrix_2 <- matrix(1:6, nrow = 2, ncol = 3,
                     byrow = TRUE)
```

```
# print the matrix
my_matrix_1
```

```
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
```

```
my_matrix_2
```

```
      [,1] [,2] [,3]
[1,]    1    2    3
[2,]    4    5    6
```

```
dim(my_matrix_1)
```

```
[1] 2 3
```

```
mat <- matrix(rnorm(n = 15, mean = 0, sd = 1),
              nrow = 5, ncol = 3)
mat <- round(mat, 2)
dim(mat)
```

```
[1] 5 3
```

```
ncol(mat)
```

```
[1] 3
```

```
nrow(mat)
```

```
[1] 5
```

```
# coerce the matrix to a vector  
# 2D to 1D  
vect <- as.vector(mat)  
vect
```

```
[1] -1.04  0.82  1.00  0.43  0.47 -1.10  0.61  0.02  0.74  1.25 -0.73 -0.07  
[13]  1.03  0.82 -0.45
```

```
length(vect)
```

```
[1] 15
```

```
mat
```

```
      [,1] [,2] [,3]  
[1,] -1.04 -1.10 -0.73  
[2,]  0.82  0.61 -0.07  
[3,]  1.00  0.02  1.03  
[4,]  0.43  0.74  0.82  
[5,]  0.47  1.25 -0.45
```

```
mat[2, ]
```

```
[1]  0.82  0.61 -0.07
```

```
mat[5, c(1, 3)]
```

```
[1]  0.47 -0.45
```

Arrays

```
# create a 2-dimensional array  
my_array <- array(1:27, dim = c(3, 3, 3))
```

```
# print the array  
my_array
```

```
, , 1  
      [,1] [,2] [,3]  
[1,]    1    4    7  
[2,]    2    5    8  
[3,]    3    6    9  
  
, , 2  
      [,1] [,2] [,3]  
[1,]   10   13   16  
[2,]   11   14   17  
[3,]   12   15   18  
  
, , 3  
      [,1] [,2] [,3]  
[1,]   19   22   25  
[2,]   20   23   26  
[3,]   21   24   27
```

```
dim(my_array)
```

```
[1] 3 3 3
```

```
my_array[2, 2, 2]
```

```
[1] 14
```

```
# square bracket notation:  
# [row, column, depth]  
my_array[2, 3, 3]
```

```
[1] 26
```

```
as.vector(my_array)
```



```
[1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
25
[26] 26 27
```

Making a data.frame

```
# Create a vector of dates
dates <- as.Date(c("2022-01-01", "2022-01-02", "2022-01-03",
                  "2022-01-04", "2022-01-05"))

# Create a vector of numeric data
numeric_data <- rnorm(n = 5, mean = 0, sd = 1)

# Create a vector of categorical data
categorical_data <- c("A", "B", "C", "A", "B")

# Combine the vectors into a data.frame
my_dataframe <- data.frame(day = dates,
                           number = numeric_data,
                           category = categorical_data)

# Print the dataframe
my_dataframe
```

	day	number	category
1	2022-01-01	1.1059215	A
2	2022-01-02	1.2593616	B
3	2022-01-03	-0.6061951	C
4	2022-01-04	-1.4083780	A
5	2022-01-05	1.6957718	B

```
class(my_dataframe)
```

```
[1] "data.frame"
```

```
colnames(my_dataframe)
```

```
[1] "day"      "number"   "category"
```

```
rownames(my_dataframe)
```

```
[1] "1" "2" "3" "4" "5"
```

```
dim(my_dataframe)
```

```
[1] 5 3
```

```
str(my_dataframe)
```

```
'data.frame':  5 obs. of  3 variables:
 $ day      : Date, format: "2022-01-01" "2022-01-02" ...
 $ number   : num  1.106 1.259 -0.606 -1.408 1.696
 $ category : chr   "A" "B" "C" "A" ...
```

```
mean(my_dataframe$number)
```

```
[1] 0.4092964
```

```
range(my_dataframe$day)
```

```
[1] "2022-01-01" "2022-01-05"
```

```
summary(my_dataframe)
```

	day	number	category
Min.	:2022-01-01	Min. : -1.4084	Length:5
1st Qu.	:2022-01-02	1st Qu.: -0.6062	Class :character
Median	:2022-01-03	Median : 1.1059	Mode :character
Mean	:2022-01-03	Mean : 0.4093	
3rd Qu.	:2022-01-04	3rd Qu.: 1.2594	
Max.	:2022-01-05	Max. : 1.6958	

```
glimpse(my_dataframe)
```

```
Rows: 5
Columns: 3
$ day      <date> 2022-01-01, 2022-01-02, 2022-01-03, 2022-01-04, 2022-01-05
```

```
$ number    <dbl> 1.1059215, 1.2593616, -0.6061951, -1.4083780, 1.6957718
$ category  <chr> "A", "B", "C", "A", "B"
```

```
my_tibble <- as_tibble(my_dataframe)
class(my_tibble)
```

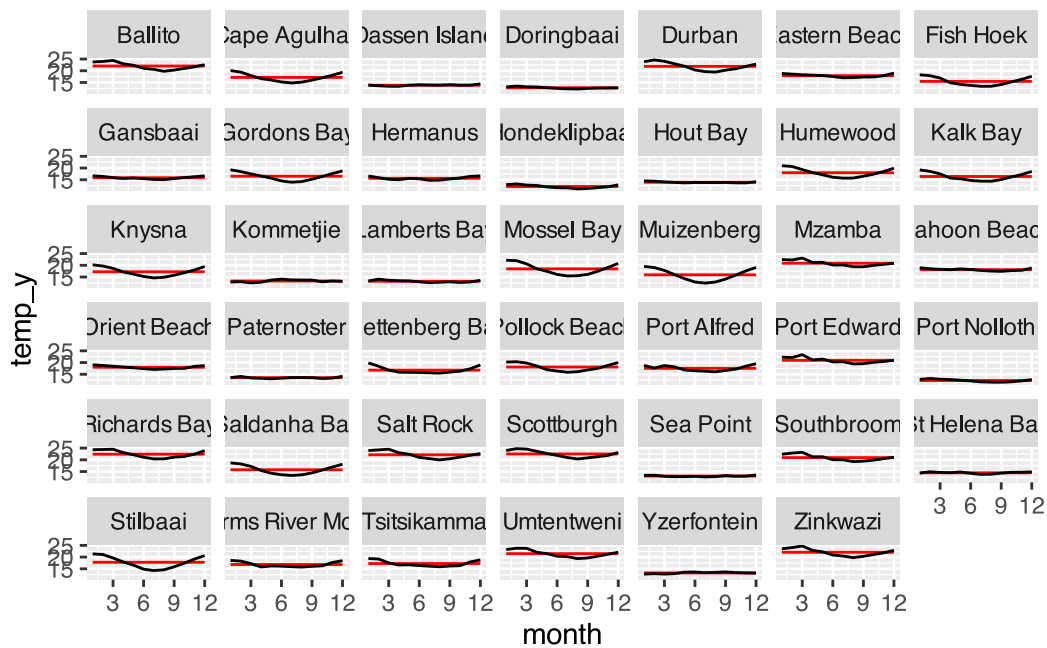
```
[1] "tbl_df"      "tbl"        "data.frame"
```

```
my_tibble
```

```
# A tibble: 5 × 3
  day          number category
<date>      <dbl> <chr>
1 2022-01-01  1.11  A
2 2022-01-02  1.26  B
3 2022-01-03 -0.606 C
4 2022-01-04 -1.41  A
5 2022-01-05  1.70  B
```

Exercise in frustration

```
read_csv("../data/SAMOS/other_data/SACTN_SAWS.csv") |>
  mutate(month = as.integer(month(date, label = FALSE))) |>
  group_by(site) |>
  mutate(temp_y = mean(temp, na.rm = TRUE)) |>
  group_by(site, month) |>
  summarise(temp_m = mean(temp, na.rm = TRUE),
            temp_y = mean(temp_y)) |>
  ggplot(aes(x = month)) +
    geom_line(aes(y = temp_y), colour = "red") +
    geom_line(aes(y = temp_m)) +
    scale_x_continuous(breaks = c(3, 6, 9, 12)) +
    facet_wrap(~site)
```



Reading in data into R

For this exercise, we will use the `laminaria.csv` file. It is located in the `data/SAMOS/other_data` folder.

```
laminaria <- read_csv("../data/SAMOS/other_data/laminaria.csv")
```

```
laminaria
```

```
# A tibble: 140 × 12
  region site      Ind blade_weight blade_length blade_thickness
  <dbl> <chr> <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 WC    Kommetjie  2        1.9        160         2
1.5
2 WC    Kommetjie  3        1.5        120        1.4
2.25
3 WC    Kommetjie  4        0.55       110        1.5
1.15
4 WC    Kommetjie  5        1         159        1.5
2.6
5 WC    Kommetjie  6        2.3       149         2
NA
6 WC    Kommetjie  7        1.6       107        1.75
```

```

2.9
 7 WC      Kommetjie      8      0.65      104      2
0.75
 8 WC      Kommetjie     10      0.95      111     1.25
1.6
 9 WC      Kommetjie     11      2.3       178     2.5
4.2
10 FB      Bordjiesti...   1      1.75      145      1
0.75
# i 130 more rows
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

```
head(laminaria, 7)
```

```

# A tibble: 7 × 12
  region site      Ind blade_weight blade_length blade_thickness stipe_mass
  <chr>  <chr>    <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 WC    Kommetjie    2        1.9        160         2        1.5
2 WC    Kommetjie    3        1.5        120         1.4       2.25
3 WC    Kommetjie    4        0.55       110         1.5       1.15
4 WC    Kommetjie    5         1        159         1.5       2.6
5 WC    Kommetjie    6        2.3        149         2        NA
6 WC    Kommetjie    7        1.6        107         1.75      2.9
7 WC    Kommetjie    8        0.65       104         2        0.75
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

```
tail(laminaria, 2)
```

```

# A tibble: 2 × 12
  region site      Ind blade_weight blade_length blade_thickness stipe_mass
  <chr>  <chr>    <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 WC    Rocky Bank   12        2.1        194         1.4       3.75
2 WC    Rocky Bank   13        1.3        160         1.9       2.45
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

```
summary(laminaria)
```

region	site	Ind	blade_weight
Length:140	Length:140	Min. : 1.000	Min. :0.125
Class :character	Class :character	1st Qu.: 4.000	1st Qu.:1.130

```

Mode :character  Mode :character  Median : 6.500  Median :1.700
Mean  : 6.586  Mean  :1.700
3rd Qu.:10.000  3rd Qu.:2.250
Max.   :13.000  Max.   :3.800

blade_length  blade_thickness  stipe_mass  stipe_length
Min.   :100.0  Min.   :0.100  Min.   :0.1250  Min.   : 34.0
1st Qu.:121.0  1st Qu.:1.200  1st Qu.:0.6325  1st Qu.: 85.0
Median :136.0  Median :1.700  Median :1.1750  Median :109.0
Mean   :139.1  Mean   :2.666  Mean   :1.3511  Mean   :111.3
3rd Qu.:156.2  3rd Qu.:4.100  3rd Qu.:1.6875  3rd Qu.:134.5
Max.   :205.0  Max.   :9.100  Max.   :5.6000  Max.   :224.0
NA's    :2

stipe_diameter  digits  thallus_mass  total_length
Min.   : 2.20  Min.   : 7.00  Min.   : 337  Min.   :145.0
1st Qu.:35.88  1st Qu.:12.00  1st Qu.:1750  1st Qu.:209.0
Median :42.00  Median :15.00  Median :2650  Median :246.5
Mean   :39.21  Mean   :15.77  Mean   :2929  Mean   :249.8
3rd Qu.:50.00  3rd Qu.:19.00  3rd Qu.:3662  3rd Qu.:283.2
Max.   :76.00  Max.   :29.00  Max.   :9400  Max.   :443.0

```

```
glimpse(laminaria)
```

```

Rows: 140
Columns: 12
$ region      <chr> "WC", "WC", "WC", "WC", "WC", "WC", "WC", "WC", "WC",
...
$ site        <chr> "Kommetjie", "Kommetjie", "Kommetjie", "Kommetjie",
"K...
$ Ind         <dbl> 2, 3, 4, 5, 6, 7, 8, 10, 11, 1, 3, 4, 5, 6, 7, 8, 9,
1...
$ blade_weight <dbl> 1.90, 1.50, 0.55, 1.00, 2.30, 1.60, 0.65, 0.95, 2.30,
...
$ blade_length <dbl> 160, 120, 110, 159, 149, 107, 104, 111, 178, 145,
146,...
$ blade_thickness <dbl> 2.00, 1.40, 1.50, 1.50, 2.00, 1.75, 2.00, 1.25, 2.50,
...
$ stipe_mass   <dbl> 1.50, 2.25, 1.15, 2.60, NA, 2.90, 0.75, 1.60, 4.20,
0...
$ stipe_length <dbl> 120, 149, 97, 167, 146, 161, 110, 136, 176, 82, 118,
1...
$ stipe_diameter <dbl> 56.0, 68.5, 69.0, 60.0, 73.0, 63.0, 51.0, 56.0, 76.0,
...
$ digits       <dbl> 12, 12, 13, 8, 15, 17, 11, 11, 8, 19, 20, 23, 20, 24,
...

```

```
$ thallus_mass    <dbl> 3000, 3750, 1700, 3600, 5100, 4500, 1400, 2550, 6500,
...
$ total_length    <dbl> 256, 269, 207, 326, 295, 268, 214, 247, 354, 227,
264,...
```

```
names(laminaria)
```

```
[1] "region"      "site"        "Ind"         "blade_weight"
[5] "blade_length" "blade_thickness" "stipe_mass"  "stipe_length"
[9] "stipe_diameter" "digits"      "thallus_mass" "total_length"
```

```
library(skimr)
skim(laminaria) # using built-in `iris` data
```

```
Name                laminaria
Number of rows      140
Number of columns   12
```

Column type frequency:

```
character           2
numeric             10
```

```
Group variables      None
```

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
region	0	1	2	2	0	2	0
site	0	1	7	17	0	13	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Ind	0	1.00	6.59	3.52	1.00	4.00	6.50	10.00	13.0	
blade_weight	0	1.00	1.70	0.77	0.12	1.13	1.70	2.25	3.8	
blade_length	0	1.00	139.06	23.88	100.00	121.00	136.00	156.25	205.0	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
blade_thickness	0	1.00	2.67	2.14	0.10	1.20	1.70	4.10	9.1	
stipe_mass	2	0.99	1.35	0.99	0.12	0.63	1.17	1.69	5.6	
stipe_length	0	1.00	111.26	39.32	34.00	85.00	109.00	134.50	224.0	
stipe_diameter	0	1.00	39.21	17.97	2.20	35.88	42.00	50.00	76.0	
digits	0	1.00	15.77	4.80	7.00	12.00	15.00	19.00	29.0	
thallus_mass	0	1.00	2928.56	1558.51	337.00	1750.00	2650.00	3662.50	9400.0	
total_length	0	1.00	249.81	55.30	145.00	209.00	246.50	283.25	443.0	

Tidyverse sneak peak

```
# Note: output NOT assigned to an object (no `<-`):
laminaria %>% # Tell R which dataframe you are using ("%>%" == "|>" )
  select(site, total_length) # Select only specific columns
```

```
# A tibble: 140 × 2
  site          total_length
<chr>          <dbl>
1 Kommetjie      256
2 Kommetjie      269
3 Kommetjie      207
4 Kommetjie      326
5 Kommetjie      295
6 Kommetjie      268
7 Kommetjie      214
8 Kommetjie      247
9 Kommetjie      354
10 Bordjiestif North  227
# i 130 more rows
```

```
laminaria %>%
  select(site, total_length) %>% # Select specific columns first
  slice(56:78)
```

```
# A tibble: 23 × 2
  site          total_length
<chr>          <dbl>
1 Miller's Point    215
2 Miller's Point    186
3 Miller's Point    179
4 Miller's Point    268
```



```

5 Miller's Point      194
6 Miller's Point      207
7 Miller's Point      167
8 Baboon Rock         146
9 Baboon Rock         147
10 Baboon Rock         190
# i 13 more rows

```

what does the '56:78' do? Change some numbers and run the code again. What happens?

```

laminaria %>%
  filter(site == "Kommetjie")

```

```

# A tibble: 9 × 12
  region site      Ind blade_weight blade_length blade_thickness stipe_mass
  <chr> <chr>    <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 WC    Kommetjie  2        1.9        160         2        1.5
2 WC    Kommetjie  3        1.5        120         1.4       2.25
3 WC    Kommetjie  4        0.55       110         1.5       1.15
4 WC    Kommetjie  5        1          159         1.5       2.6
5 WC    Kommetjie  6        2.3        149         2         NA
6 WC    Kommetjie  7        1.6        107         1.75      2.9
7 WC    Kommetjie  8        0.65       104         2         0.75
8 WC    Kommetjie 10        0.95       111         1.25      1.6
9 WC    Kommetjie 11        2.3        178         2.5       4.2
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

```

laminaria |>
  filter(blade_weight >= 1.5)

```

```

# A tibble: 83 × 12
  region site      Ind blade_weight blade_length blade_thickness
  stipe_mass
  <chr> <chr>    <dbl>      <dbl>      <dbl>      <dbl>
<dbl>
1 WC    Kommetjie  2        1.9        160         2
1.5
2 WC    Kommetjie  3        1.5        120         1.4
2.25
3 WC    Kommetjie  6        2.3        149         2
NA
4 WC    Kommetjie  7        1.6        107         1.75

```

```

2.9
5 WC      Kommetjie      11      2.3      178      2.5
4.2
6 FB      Bordjiesti...  1      1.75     145      1
0.75
7 FB      Bordjiesti...  3      2.35     146      1.75
1.25
8 FB      Bordjiesti...  4      1.7      116      1.25
NA
9 FB      Bordjiesti...  6      2.45     136      1
1.95
10 FB     Bordjiesti...  7      2        124      1.25
1.3
# i 73 more rows
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

```

lam_kom <- laminaria %>%
  filter(site == "Kommetjie")

```

```

laminaria %>% # Tell R which dataset to use
  filter(site == "Kommetjie") %>% # Filter out only records from Kommetjie
  nrow() # Count the number of remaining rows

```

```
[1] 9
```

```

laminaria %>% # Tell R which dataset to use
  filter(total_length == max(total_length)) # Select row with max total length

```

```

# A tibble: 1 × 12
  region site      Ind blade_weight blade_length blade_thickness
  <chr> <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 WC    Olifantsbos    4        3.35        205         1
3
# i 5 more variables: stipe_length <dbl>, stipe_diameter <dbl>, digits <dbl>,
#   thallus_mass <dbl>, total_length <dbl>

```

Some new functions: summarise, mutate, and group_by

```

laminaria %>% # Chose the dataframe
  summarise(avg_bld_len = mean(blade_length)) # Calculate mean blade length

```

```
# A tibble: 1 × 1
  avg_bld_len
    <dbl>
1      139.
```

```
laminaria %>%
  summarise(avg_stp_ln = round(mean(total_length), 1),
            sd_stp_ln = round(sd(total_length), 1))
```

```
# A tibble: 1 × 2
  avg_stp_ln sd_stp_ln
    <dbl>    <dbl>
1      250.      55.3
```

```
laminaria %>%
  summarise(avg_stp_ms = mean(stipe_mass, na.rm = TRUE))
```

```
# A tibble: 1 × 1
  avg_stp_ms
    <dbl>
1      1.35
```

```
res <- laminaria %>%
  group_by(site) %>%
  summarise(var_bl = var(blade_length),
            n_bl = n(),
            sd_bl = sd(blade_length)) %>%
  mutate(se_bl = var_bl / sqrt(n_bl))
```

```
res
```

```
# A tibble: 13 × 5
  site          var_bl n_bl sd_bl se_bl
  <chr>        <dbl> <int> <dbl> <dbl>
1 A-Frame      342.    12  18.5  98.6
2 Baboon Rock  360.    12  19.0 104.
3 Batsata Rock 655.    10  25.6 207.
4 Betty's Bay  487.    12  22.1 141.
5 Bordjiestif North 283.    10  16.8 89.5
6 Buffels     207.    12  14.4 59.6
7 Buffels South 260.    12  16.1 75.1
8 Kommetjie    798.     9  28.3 266.
```

9 Miller's Point	261.	9	16.1	86.9
10 Olifantsbos	679.	10	26.1	215.
11 Rocky Bank	568.	11	23.8	171.
12 Roman Rock	200.	11	14.1	60.3
13 Sea Point	471.	10	21.7	149.

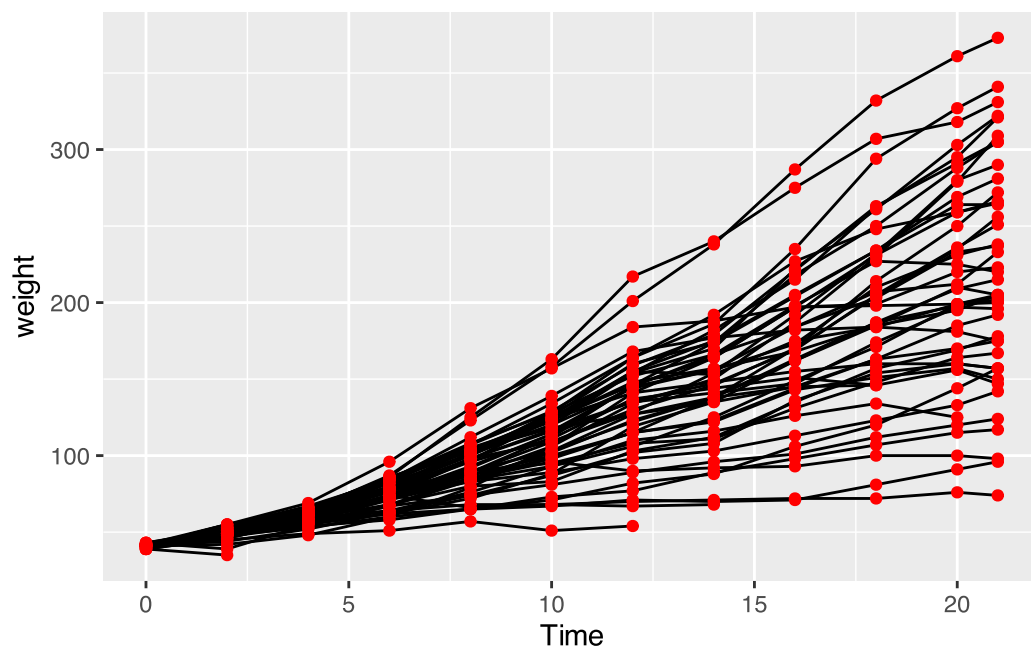
```
write.csv(res, "../data/SAMOS/other_data/laminaria_results.csv")
```

Graphics with R

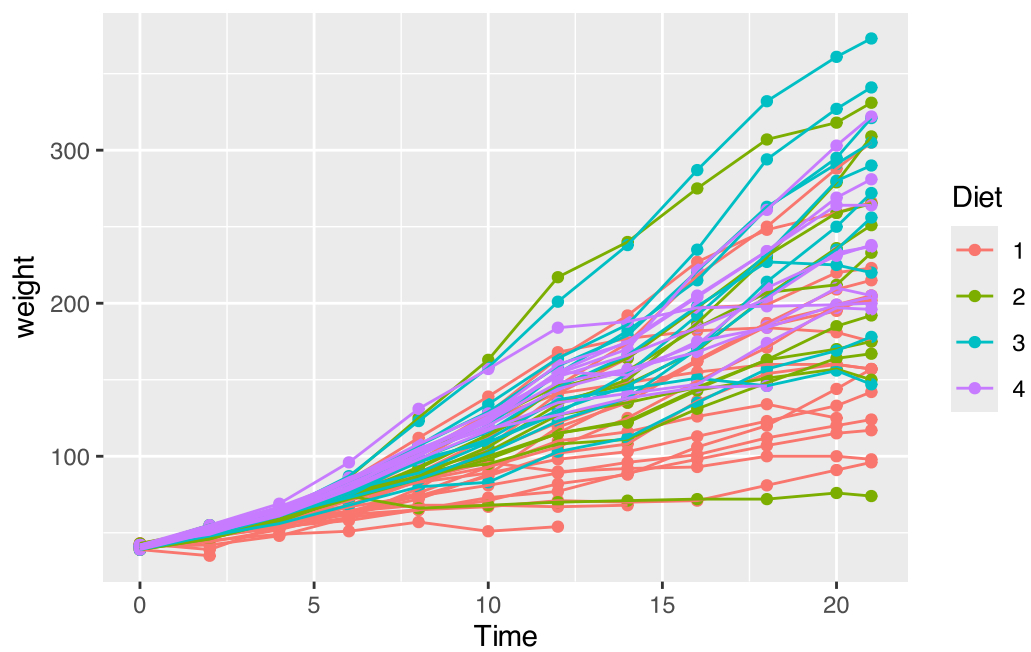
```
data()
```

```
# Load data
ChickWeight <- datasets::ChickWeight

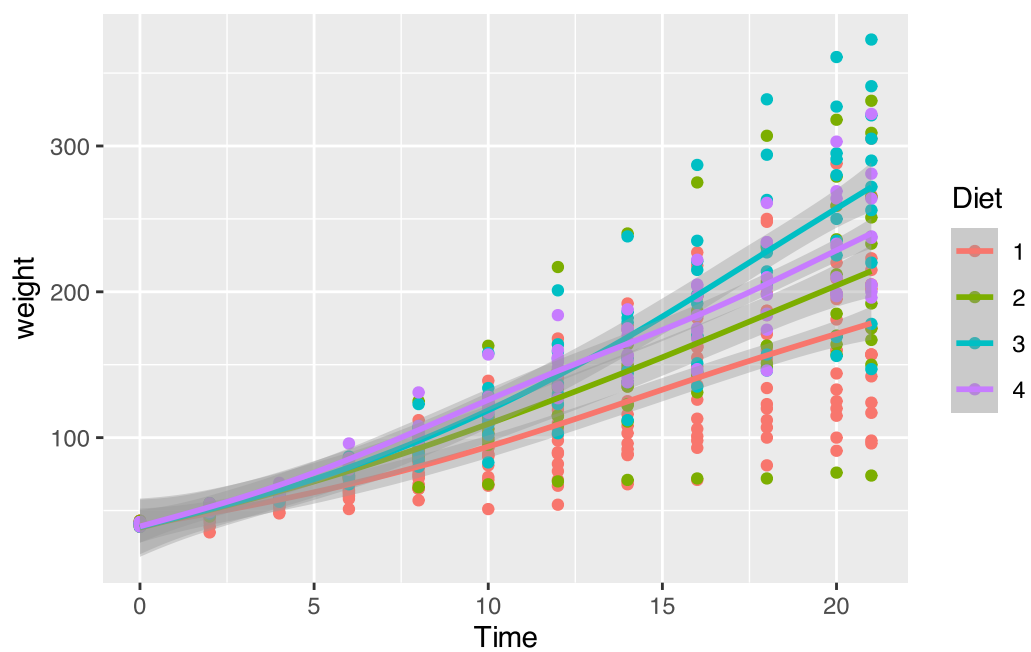
# Create a basic figure
ggplot(data = ChickWeight, aes(x = Time, y = weight)) +
  geom_line(aes(group = Chick)) +
  geom_point(colour = "red")
```



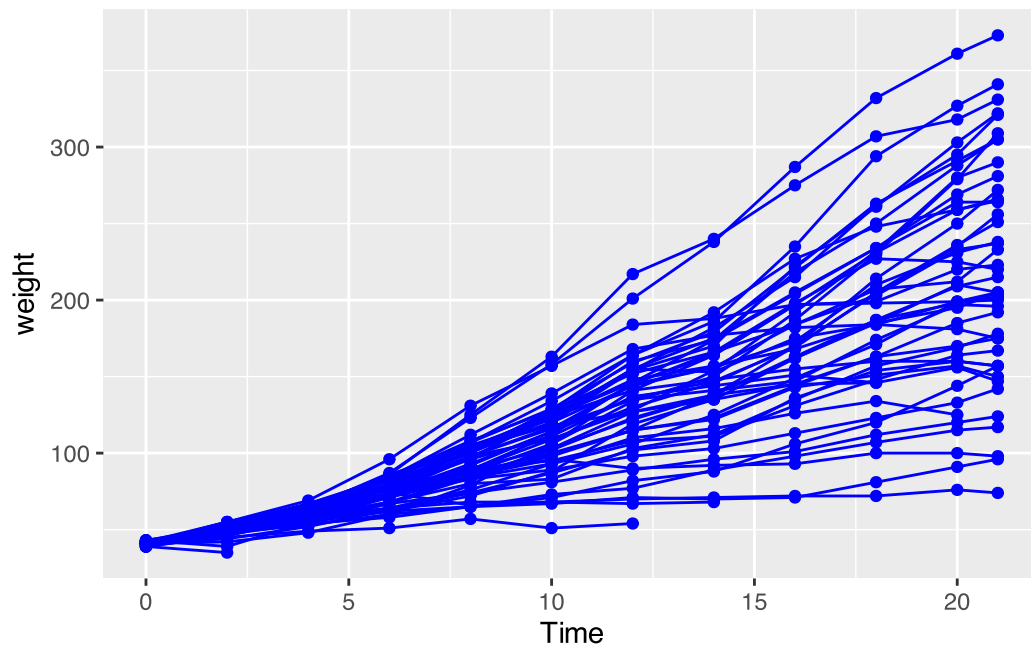
```
ggplot(data = ChickWeight, aes(x = Time, y = weight, colour = Diet)) +
  geom_point() +
  geom_line(aes(group = Chick))
```



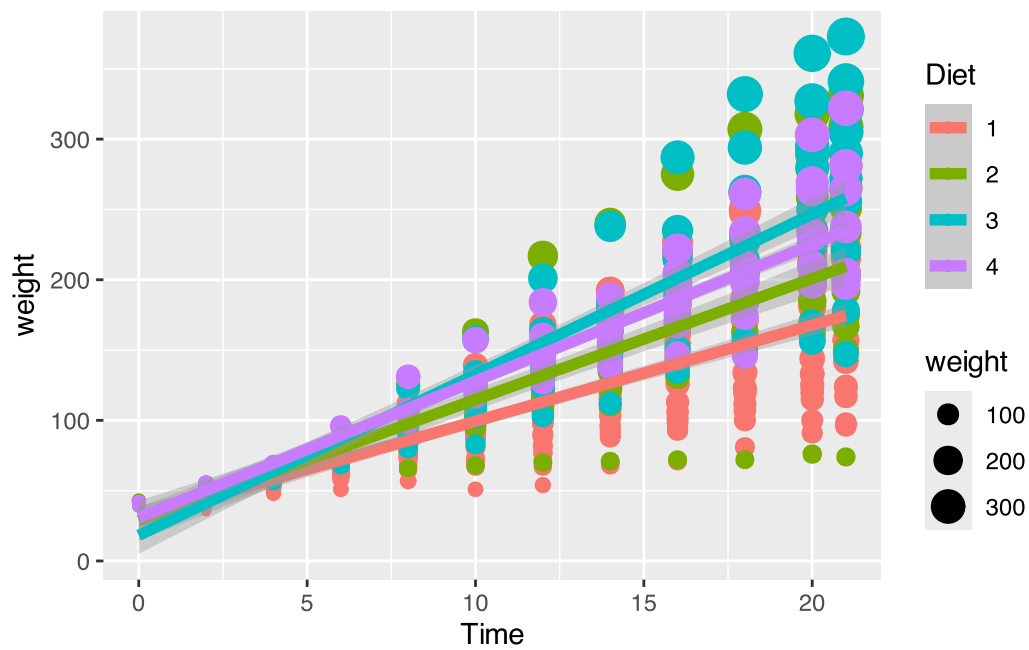
```
ggplot(data = ChickWeight, aes(x = Time, y = weight, colour = Diet)) +
  geom_point() +
  geom_smooth(method = "gam")
```



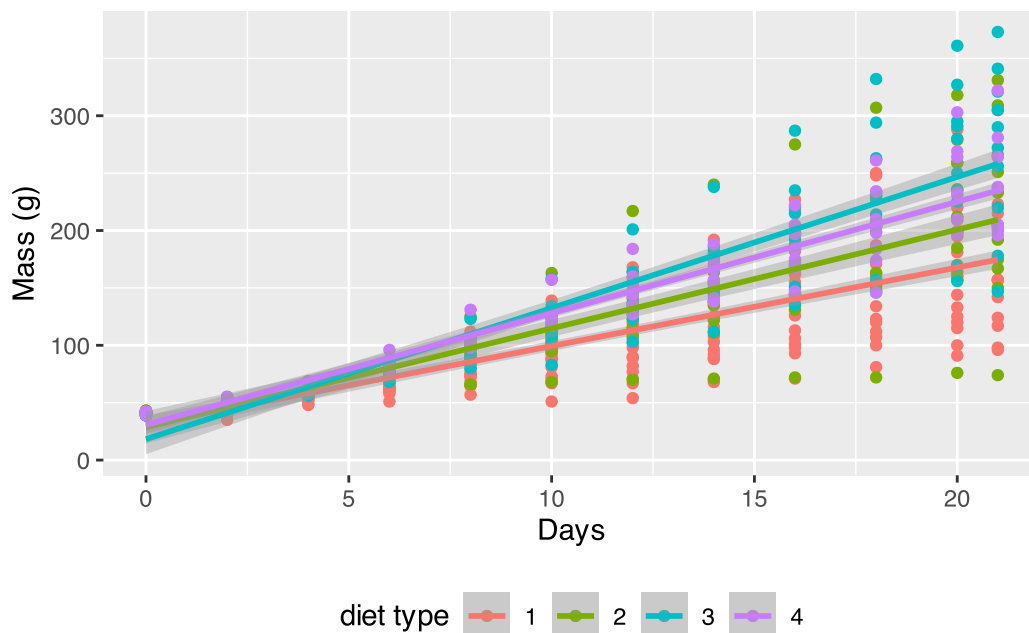
```
ggplot(data = ChickWeight, aes(x = Time, y = weight)) +
  geom_point(colour = "blue") +
  geom_line(aes(group = Chick), colour = "blue")
```



```
ggplot(data = ChickWeight, aes(x = Time, y = weight, colour = Diet)) +
  geom_point(aes(size = weight)) +
  geom_smooth(method = "lm", linewidth = 2.0)
```



```
ggplot(data = ChickWeight, aes(x = Time, y = weight, colour = Diet)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(x = "Days", y = "Mass (g)", colour = "diet type") + # Change the labels
  theme(legend.position = "bottom") # Change the legend position
```



Bibliography